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Effectiveness of Natural Language Processing Method Using Inductive Learning

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ABSTRACT
A technology of natural language processing is higher recently. However the traditional technology of it can not deal with a practical situation. Because there are various language phenomena of the practical situation. Therefore we propose an inductive learning method for natural language processing. This method is robust. Because the system based on it can adapt to a user dynamically by acquiring rules from examples using the inductive learning. We applied this method to four kinds of data. And we have developed four applications. We have carried out the performance evaluation experiments for them. We discuss the effectiveness of our method for natural language processing using these results.

KEY WORDS
inductive learning, natural language processing, machine translation, spoken dialogue processing, morphological analysis

1. Introduction

The main method of natural language processing was an analytical approach [8]. In this method, a system based on it resolves problems using the prepared knowledge on no ambiguous situation. This method is also called the Knowledge-based approach. However this approach is not robust. The system based on this approach can deal with a limited situation that a developer expects. However there are many situations which the developer can not expect in the world which natural language can express. Therefore this approach can not deal with many varieties of language phenomena. In the 1990s, the Corpus-based approach [4] and the Statistics-based approach [5] were proposed in order to resolve this problem. A method based on this approach is robust. However the method needs a huge corpus in order to increase the precision of the system based on this method. The development of a huge corpus needs a heavy labor of a developer.

We have proposed a method of natural language processing using inductive learning[3] in order to resolve this problem. We call it Natural Language Processing Method Using Inductive Learning(IL-NLP). The IL-NLP is robust and does not need a huge corpus. The method using a probability in the statistics approach is general. However it is not suitable for a specific context. Therefore, in this approach, the precision of the system based on it is low for the other data except the learning data. It is important for natural language processing that the system can get the solution that is suitable for the specific context by adapting to the target dynamically. The system based on the IL-NLP is the most desirable that it can adapt to a target by adapting a rule dynamically using inductive learning, even if the target changes. Moreover it does not need a huge corpus because it can acquire a needed rule from a comparative few examples using the inductive learning. Therefore it is the most desirable for natural language processing.

In this paper, we describe the purpose of our research, the results of applications and the consideration of effectiveness of the IL-NLP. At last, we describe the conclusion and a future problem for it.

2. Goal of our research

We are interested in native capability for acquisition of language and knowledge of a human being. And the final goal of our research is that "Practical realization of native capability for acquisition of language and knowledge of a human being on engineering". We need to make its mechanism clear in order to achieve this final goal. The system that has language and knowledge acquisition capability can learn various kinds of information from an environment and grow up. In informational society, there are many problems such that the most of systems are very complex and difficult to use them, and we need heavy labor for developing a lot of software. We could resolve these problems if we could develop this system that has language and knowledge acquisition capability. A baby can not speak language and has no knowledge. However it grows into an adult which can speak language very well and has various kinds of knowledge by learning from the environment. Therefore a human being has native capability for acquisi-

tion of language and knowledge certainly.

There are already some researches on the realization of native capability for acquisition of language and knowledge [12]. However the systems based on them are not effective on engineering. And there are many researches on them in psychology [6]. These researches are effective from point of view how a baby acquires the language. However they do not deal with problem how to develop this acquiring capability on the engineering.

3. Basic idea

We assume that the native capability is to judge which two things are the same or not. And then we developed the system using the IL-NLP based on this assumption and carried out the experiments in order to evaluate the effectiveness of it. It needs to acquire correspondences of two kinds of character strings by only native capability in order to achieve the goal of our research. By the way, how does a human being determine these correspondences? Table 1 shows an example of extraction of correspondences from unknown character strings by the human being.

Table 1. An example of extraction of correspondences from unknown character strings.

Input1	$\alpha\ \theta\ \sigma\ \underline{\psi\ \delta}\ \lambda\ \vartheta$	
Input2	$\Xi\ \Sigma\ \underline{\psi\ \delta}\ \Upsilon\ \Phi\ \Theta$	
Segment1	$\alpha\ \theta\ \sigma$	$\Xi\ \Sigma$
Segment2	$\psi\ \delta$	$\psi\ \delta$
Segment3	$\lambda\ \vartheta$	$\Upsilon\ \Phi\ \Theta$

The human being finds out common parts between two character strings when it watches these unknown character strings. In the example of Table 1, we consider that it watches underlined ($\psi\quad\delta$), and then makes different parts correspond in order of their appearance. As a result, it acquires segment 1, segment 2 and segment 3 in Table 1. This extraction process is consistent with our assumption that native capability is to judge which two things are same or not. However it can also determine the reverse correspondences about the different parts. The correspondence is the reverse order of appearance. It sometimes considers this correspondence. Therefore heuristics which depends on the target determine which the correspondence is the order of appearance or not. It is important to consider how to acquire this heuristics. However this problem is very difficult. Therefore this heuristics is given beforehand in this paper and we will describe the consideration about this mechanism in another paper.

Moreover primitives are extracted from the segments extracted as mentioned above. The primitives are acquired by extracting common parts and different parts between the segments. Table 2 shows an example of the primitives extracted from the segments. In Table 2, the common part

Table 2. An example of primitives extracted from segments.

Segment 1	$\alpha\ \underline{\theta\ \sigma}$	$\Upsilon\ \underline{\Phi\ \Theta}$
Segment 2	$\underline{\theta\ \sigma}\ \gamma\ \mu$	$\underline{\Phi\ \Theta}\ \Sigma$
Primitive 1	α	Υ
Primitive 2	$\theta\ \sigma$	$\Phi\ \Theta$
Primitive 3	$\gamma\ \mu$	Σ

($\theta\ \sigma, \Phi\ \Theta$) between the segment 1 and the segment 2 is extracted as the primitive 2. And the different parts on both sides of them are extracted as the primitive 1 and the primitive 3 respectively. In this way, these three primitives are extracted by separating the common part and the different part. Two sources of segment 1 and segment 2 do not need when there are these three primitives. Because they can be generated from these three primitives. This extraction process is based on our assumption that the native capability is to judge which two things are the same or not. However the heuristics mentioned above is also used for the generation of primitives. The method which knowledge is acquired by extracting common parts and different parts in multistage steps are a kind of the inductive learning. The learning process of the IL-NLP is based on these processes as mentioned above.

4. Applications of IL-NLP

The final goal of our research is "Practical realization of native capability for acquisition of language and knowledge of a human being on engineering." Concretely, we have proposed four applications using IL-NLP for four kinds of data as mentioned below. Fist, we use a Japanese Kanji-Kana mixed sentence. This sentence is one kind of a character string. We proposed a morphological analysis method using inductive learning[1]. Second, we use a pair of a Japanese Kanji-Kana mixed sentence and its Kana sentence expressed its pronunciation. They are two kinds of character strings. They are the same on syntax, but different on the surface expression. We proposed a Japanese Kana-Kanji translation method using inductive learning[2]. Third, we use a pair of a source sentence and its target sentence in machine translation. In these sentences, they are the same on semantics, but different on syntax. We proposed a method of the machine translation using inductive learning with genetic algorithms[7]. Fourth, we use a speech on spoken dialogue. The speech consists of a pair of a user's utterance and the system's reply for it. In these utterances, they are the same on syntax, but they are different on semantics. However, there is the relation of cause and effect between them on semantics. We propose a method of spoken dialogue processing using inductive learning with genetic algorithms[10].

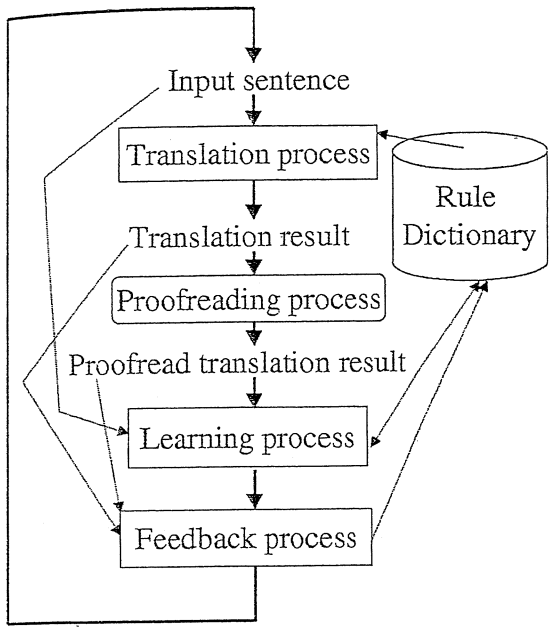


Figure 1. Process of IL-NLP.

Figure 1 shows the process of the IL-NLP. It consists of four processes that are the translation process, the proofreading process, the learning process and the feedback process. When a sentence is inputted, the system based on the IL-NLP translates it using the rules. The rules were acquired from examples in the learning process using the inductive learning. If there are some errors in the translated result, a user needs to correct them. In the learning process, the system acquires rules from the pairs of an inputted sentence and its proofread result using the inductive learning. At last, the system increases the degrees of priorities of the rules that are used for the correct translation, and decreases the degrees of priorities of the rules that are used for the erroneous translation in the feedback process. This method acquired rules from Kanji-Kana mixed sentences in the morphological analysis, the pairs of a pronounced sentence and its Kanji-Kana mixed sentence in Kana-Kanji translation, the pairs of a source sentence and its translated sentence in the machine translation, and the pairs of a user's utterance and the system's reply for it in the spoken dialogue processing. And the IL-NLP uses not only the inductive learning but also genetic algorithms in the applications of the machine translation and the spoken dialogue processing. Because the performance of the inductive learning is not enough for them. Therefore we add the genetic algorithms to it.

We describe the detail of applications to the Kana-Kanji translation and the machine translation in the next two subsections and do not describe the detail of applications to the morphological analysis and the dialogue processing in this paper. Because the space is limited.

4.1 Kana-Kanji translation

We applied our method to two kinds of symbol strings that they are the same on syntax, but they are different on the surface expression. This symbol strings are a pair of a Japanese phonetic character string and its Japanese Kanji-Kana mixed sentence. We call this method "Kana-Kanji Translation Method Using Inductive Learning[3]." In this method, a normal expression and phonetic character string of a word [1] are acquired from phonetic sentence and its Kanji-Kana mixed sentence using inductive learning. And the system based on this method translates from phonetic sentence into its Kanji-Kana mixed sentence. In this process, the system uses the pairs of phonetic character strings and normal expressions of the words in order of higher certain degrees of them in the multistage steps. Their certain degrees are based on situation acquired a word and rate of correct translation of it. The processes of this method are shown in Figure 1. This method consists of three processes that are translation process, learning process and feedback learning process. The learning process is based on 3. The learning process consists of two steps. At the first step, a word candidate is acquired by common part between phonetic sentence and its proofread translation result. At the

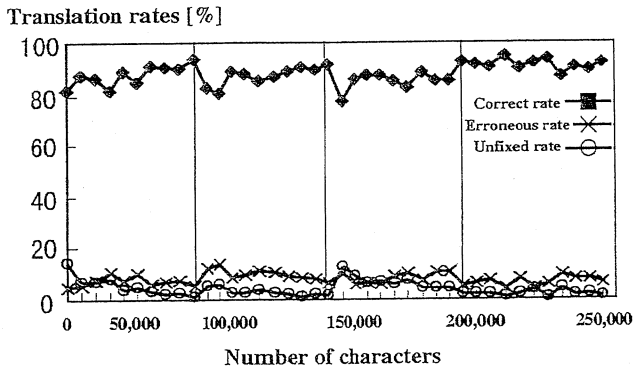


Figure 2. Changes in translation rates.

We carried out performance evaluation experiment. The initial dictionary is empty. We use the 44 papers that are concern with information engineering, mechanical engineering, applied chemistry and the humanities. Total number of characters of these data is 302,703. Figure 2 shows the changes in translation rates. Correct rate goes down slightly each changing point of data. However the correct rate recovers by learning data immediately. The correct rates of all the categories reached about 90%. These results show that the inductive learning is effective for Japanese Kana-Kanji translation.

[1] This information need for Kana-Kanji translation.

4.2 Application to Japanese input method for mobile phone

Moreover we have developed a Japanese input method for mobile phone using learning capability of this method[9]. They call this method "Number-Kanji Translation Method of Non-Segmented Japanese Sentences Using Inductive Learning with Degenerated Input." In this method, a user input Japanese sentence using 12 keys, which is keypad of telephone. Table 3[2] shows correspondences between a Kana character and a key. In this method, the degenerated input is used. Therefore this input lacks vowel information. This method recovers this lack by adapting to the user using its learning capability.

Table 3. Correspondences between Numbers and Kanas

1	*a i u e o*	2	*ka ki ku ke ko*
3	*sa shi su se so*	4	*ta chi tsu te to*
5	*na ni nu ne no*	6	*ha hi fu he ho*
7	*ma mi mu me mo*	8	*ya yu yo*
9	*ra ri ru re ro*	*	a voiced consonant
0	*wa wo n*	#	a punctuation

We carried out performance evaluation experiment. The initial dictionary is empty. We use UNIX online manual (50,000 characters) and three papers [3] (72,000 characters) on information processing. The average of correct rates of UNIX online manual and these papers are 75.1% and 76.7% respectively. These results show that inductive learning is effective for Japanese input method of mobile phone.

4.3 Machine translation

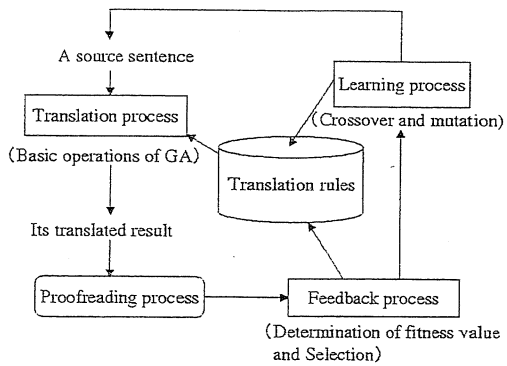


Figure 3. Process of GA-ILMT.

Next, we develop inductive learning for two kinds of symbol strings that are source sentence and its target sentence. They are similar on semantics, but are different completely on syntax. Therefore only inductive learning is not enough to learn translation rules from them. We apply genetic algorithms to inductive learning in order to strengthen it. They call this method "Machine Translation method using Inductive Learning with Genetic Algorithms" and express GA-ILMT[7]. And they express a method of inductive learning with genetic algorithms GA-IL.

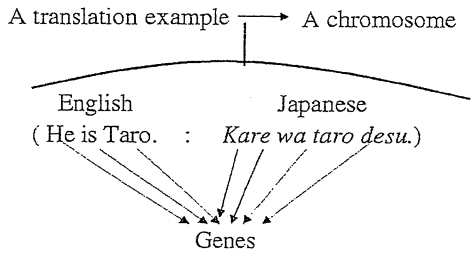


Figure 4. A chromosome and a gene

In this method, translation rules are acquired from translation examples using inductive learning. The system based on it translates from the source sentence to its target sentence using the acquired rules. Generally, a machine translation's system based on learning method needs a lot of examples. However, they are not needed in this method because the system generates translation examples from a comparatively few examples automatically using crossover of GA. Moreover algorithm of the whole system is based on GA. Therefore the system becomes optimum by selecting rules using feedback of translating information. Figure 3 shows process of GA-ILMT.

(1)The input sentence
I am your teammate.
(2)The initial group
ENGLISH JAPANESE
(I am @0. :*Watashi wa @0 desu.*)
(@0 am your @1. :*@0 wa kimi no @1 desu.*)
(@0 am @1. :*@0 wa @1 desu.*)
(3)The translation rule by crossover
ENGLISH JAPANESE
(I am your @1. :*Watashi wa kimi no @1 desu.*)
(4)The evaluation of population
ENGLISH JAPANESE
(teammate :*chiimumeito*)
(I am your @1. :*Watashi wa kimi no @1 desu.*)
(I am your teammate.
:*Watashi wa kimi no chiimumeito desu.*)
(5)The translation result
JAPANESE
Watashi wa kimi no chiimumeito desu.

Figure 5. An example of the translation process

In Figure 4, a chromosome corresponds to a translation example that consists of source sentence and target

[2] An italic character expresses Japanese.
[3] The same author wrote them.

sentence, and a gene corresponds to a word. In Figure 4, the source sentence is English, and the target sentence is Japanese. Figure 5 shows an example of the translation process. In this method, one-point crossover is used. And the rate of mutation is 2%. We carried out performance evaluation experiment for English-Japanese machine translation. We use English textbook for students of Japanese junior high school. This textbook has 1,010 translation examples (11,479 characters). And the initial dictionary is empty. The effective rate of translation is 61.9%. This result shows that inductive learning is effective in machine translation because the initial dictionary is empty.

5. Experimental results and consideration

We have carried out the performance evaluation experiment for the applications of the IL-NLP. Table 4 shows the results of applications of it. The correct rates of morphological analysis and Kana-Kanji Translation are about 90%. However the correct rates of machine translation and dialogue processing are 62% and 72% respectively. These results show that the IL-NLP is effective for the morphological analysis and the Kana-Kanji translation, and not effective in the machine translation and the dialogue processing. This reason is that the system based on the IL-NLP can determine the correspondences between the parts of an input sentences and the parts of its translated sentence in the former. Because the former is the same syntax and semantics between an input sentence and its translated sentence. However the latter is that the system can not determine the correspondences between them. Because the latter is different syntax or semantics between them.

We need to develop analysis tools of syntactic structure and semantic structure using the IL-NLP in order to resolve this problem. Because there are correspondences between an input sentence and its translated sentence on the syntactic structure and the semantic structure even in machine translation and dialogue processing. We have already developed the syntactic analysis system using the IL-NLP[11]. And we try to develop the semantic analysis system using it.

The purpose of introduction of GA is automatic generation of an example using crossover of it and the realization of feedback process by selection of it. The process of trial and error of a baby corresponds to the automatic generation of an example using the crossover. When it says an erroneous utterance, it finds that its utterance is wrong from the reaction of an adult for it. And then, it will not use these rules that were used in order to generate this erroneous utterance after it. Therefore the introduction of the GA is consistent with the purpose of this research. The correct rates of the machine translation system and the spoken dialogue system based on the IL-NLP increase about 10 points by the introduction of GA. Therefore the GA is effective for the IL-NLP from the point of view of the practical realization of the capability of language acquisition on engineering.

The development of the system based on the IL-NLP is restricted by the condition of hardware. If it is developed without restriction, the amount of processing increases extremely. Therefore we can not develop the system based on it. And a human being does not also memorize all things. It memorizes only recent things. When the condition of application of the IL-NLP is strict, a wide use does not increase. Because the generality of a rule does not increase so much. However the correct rate increases because more concrete rules are used for translation. Conversely, when the condition of application of the IL-NLP is loose, a wide use increases. Because the generality of a rule increases so much. However the correct rate decreases because more abstract rules are used for translation. The spoken dialogue system based on the IL-NLP is the former because there are not the definite general correspondences between words of a user's utterance and words of its system's reply. However the machine translation system based on the IL-NLP is the latter because there are the general correspondences between words of a source sentence and words of its translated sentence.

6. Conclusion

In this paper, we have described the effectiveness of the IL-NLP. We have developed four applications of morphological analysis method using inductive learning, Japanese Kana-Kanji translation method using inductive learning, machine translation method using the inductive learning with genetic algorithms and spoken dialogue method using it. We have carried out performance evaluation experiments for them. These results show that the IL-NLP is effective for the morphological analysis and the Kana-Kanji translation, and not so effective in the machine translation and the dialogue processing. And we describe that we develop analysis tools of syntactic structure using the IL-NLP[11] and try to develop semantic structure using the IL-NLP for resolving this problem. At last, we discuss the correct rate, the introduction of GA and the condition of development of the system based on the IL-NLP.

Future problem is the development of a semantic analysis system using the IL-NLP. We have already developed the morphological analysis method using it. Therefore a sentence could be able to translate a character string to its semantic structure only using the IL-NLP without any other knowledge if the semantic analysis system using it would be developed. Moreover we will develop many varieties of applications of natural language processing using the semantic structure into which translated a character string by the IL-NLP.

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Table 4. Results of the applications of IL-NLP.

Kind of Application	Attribution of Data	Correct rate	Heuristics
Morphological analysis	Kind: 1 Expression: same Syntax: same Semantics: same	90%	Condition of learning Promising degree evaluation Condition of word rank
Kana-Kanji translation	Kind: 2 Expression: different Syntax: same Semantics: same	90%	Condition of learning Promising degree evaluation Condition of word rank Both directional analysis
Machine translation	Kind : 2 Expression: different Syntax: different Semantics: close	62% (Without GA: 52%)	Condition of learning Promising degree evaluation Position of crossover Rate of mutation Condition of selection
Dialogue processing	Kind: 2 Expression: different Syntax: same Semantics: different	76%	Position of crossover Rate of mutation Condition of selection Condition of reply generation

ciation to the staffs of CSLI.

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